

STATISTICAL MECHANICS I

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Probability Theory

Expectation Consider a continuous random variable $X : \Omega \rightarrow \mathbb{R}$ with P.D.F $f_X(x)$. We define expectation of $Y = g(X)$ as

$$\mathbb{E}[g(X)] \equiv \langle g(X) \rangle = \int_{-\infty}^{\infty} dx g(x) f_X(x)$$

iff $\int_{-\infty}^{\infty} dx |g(x) f_X(x)| < \infty$

Characteristic Function/FT For any random variable X , its characteristic function is defined as

$$\mathcal{Q}_X(k) \equiv \tilde{f}(k) = \langle e^{-ikX} \rangle ; \forall k \in \mathbb{R}$$

$$\frac{\partial^n \tilde{f}(k)}{\partial k^n} = (-i)^n \langle X^n \rangle$$

Cumulant Generating Function CGF is defined as

$$F_C(k) = \ln \tilde{f}(k) \equiv \sum_{m=1}^{\infty} \frac{(-ik)^m}{m!} \langle X^m \rangle_c$$

$$\frac{\partial^n F_c(k)}{\partial k^n} = (-i)^n \langle X^n \rangle_c$$

Useful: $\ln(1 + \epsilon) = \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \epsilon^m$. The first 4 cumulants $\langle X^m \rangle_c$ are mean, variance, skewness, and kurtosis of the distribution respectively.

Memorylessness A random variable X is called memoryless if for any $a, b > 0$ we have $P(X > a + b | X \geq a) = P(X > b)$.

Master Equation Let us denote $\mathcal{P}(i, t)$ to be the probability of finding a random variable X in state i at time t . Then under *markovian* evolution of these states, its rate of change should only be governed by $(i - 1) \leftrightarrow i \leftrightarrow (i + 1)$. Then

$$\frac{d\mathcal{P}(i, t)}{dt} = \sum_{j \neq i} [\omega_{j \rightarrow i} \mathcal{P}(j, t) - \omega_{i \rightarrow j} \mathcal{P}(i, t)]$$

Random Walker An unbiased random walk ($p = q = 1/2$) is just a diffusion process with,

$$\frac{\partial P}{\partial t} = D \frac{\partial^2 P}{\partial x^2} ; D = \frac{(\Delta x)^2}{2\Delta t}$$

where the probability the particle is in state m , after N iterations/steps is $P_N(m) \rightarrow P(x, t)$ with $x = m\Delta x$, $t = N\Delta t$

Entropy The entropy of any probability distribution is defined as

$$\mathcal{S} = - \sum_i P(X = x_i) \ln P(X = x_i)$$

Kinetic Theory

Liouville Equation Consider a system with N particles. Let $d\Gamma \sim \prod_{j \leq N} d^3\vec{q}_j d^3\vec{p}_j$ be a region (volume) of $6N$ dimensional phase space. It follows that under hamiltonian evolution of the system, $\frac{d\Gamma}{dt} = 0$ (Shape can change, but Volume is conserved!). This implies that the probability density $f(\vec{q}_i, \vec{p}_i; t)$ (of finding the system in a particular macrostate $(\vec{q}_i, \vec{p}_i)$) doesn't change as we follow it along any trajectory in phase space. In other words,

$$\frac{df}{dt} = \{f, H\} + \frac{\partial f}{\partial t} = 0$$

At *equilibrium* $\frac{\partial f_{eq}}{\partial t} = 0$

1 -particle PDF

$$f_1(\vec{q}, \vec{p}; t) = N \int \prod_{j=2}^N d^3q_j d^3p_j f(r_1 = \vec{r}, r_2, \dots, r_N, \vec{p}_1 = \vec{p}, \vec{p}_2, \dots, \vec{p}_N)$$

BBGKY Hierarchy It states the evolution of 1 particle PDF is dependent on streaming term (dominant in *dilute gas*) and collision term (which depends on f_2)

$$\frac{\partial f_1}{\partial t} = \{H_1, f_1\} + \left(\frac{\partial f_1}{\partial t} \right)_{coll}$$

Boltzmann Equation Consider 2 particles colliding at $\vec{q}$ with momenta $\vec{p}_1, \vec{p}_2$ and emerging with $\vec{p}_1', \vec{p}_2'$. Then under the assumption of *molecular chaos* ($f_2(\vec{q}, \vec{q}, \vec{p}_1, \vec{p}_2) = f_1(\vec{q}, \vec{p}_1) f_1(\vec{q}, \vec{p}_2)$), the collision integral becomes

$$\left(\frac{\partial f_1}{\partial t} \right)_{coll} = \int d^3p_2 d^3p_1' d^3p_2' \omega(\vec{p}_1', \vec{p}_2' | \vec{p}_1, \vec{p}_2) \times [f_1(\vec{q}, \vec{p}_1') f_1(\vec{q}, \vec{p}_2') - f_1(\vec{q}, \vec{p}_1) f_1(\vec{q}, \vec{p}_2)]$$

$\mathcal{H}$ -Theorem Consider the quantity

$$\mathcal{H}(t) \equiv \int d^3q d^3p f_1(\vec{q}, \vec{p}; t) \ln f_1(\vec{q}, \vec{p}; t)$$

From our definition of entropy, it is clear that

$$\mathcal{S} = -k_B \mathcal{H}$$

The theorem states that

$$\frac{d\mathcal{H}}{dt} \leq 0 \iff \frac{d\mathcal{S}}{dt} \geq 0$$

Microcanonical Ensemble

Construction In an isolated system with no work/heat/particle exchange, all the microstates (μ) corresponding to a particular *macrostate* $\mathbf{M}(E, \vec{x})$ all lie on the surface $H(\mu) = E$ in phase space with probability $p_{(E, \vec{x})}(\mu) = 1/\Omega(E, \vec{x})$.

- entropy is given by $\mathcal{S} = k_B \ln(\Omega(E, V, N))$
- Use the 1st Law: $dE = Td\mathcal{S} - PdV + \mu dN$ to deduce relations between p, μ, T with $\mathcal{S}$.
- Heat Capacity as $C_V = \left. \frac{\partial E}{\partial T} \right|_{N, V}$
- $C_V = T \left(\frac{\partial \mathcal{S}}{\partial T} \right)_{V, N}$; $C_P = T \left(\frac{\partial \mathcal{S}}{\partial T} \right)_{P, N}$; $\gamma = C_P/C_V$
- Compressibility $\kappa = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T$
- Bulk Modulus $B = -V \left(\frac{\partial P}{\partial V} \right)_T$
- Temperature $\frac{1}{T} = \left(\frac{\partial \mathcal{S}}{\partial E} \right)_{V, N}$
- Pressure $\left(\frac{\partial \mathcal{S}}{\partial V} \right)_{E, N} \equiv p/T$

In general, the total number of possible microstates for N identical and indistinguishable particles is given as

$$\Omega(E, V, N) \equiv \int d\Gamma_{id} = \frac{1}{h^{3N} N!} \int_{H=E} \prod_{i \leq N} d^3\vec{q}_i d^3\vec{p}_i$$

Classical Ideal Gas

$$\int \prod_{i \leq N} d^3\vec{q}_i = V^N$$

$$\int_{\sum_i \vec{p}_i^2 = 2mE} \prod_{i \leq N} d^3\vec{p}_i = A_{3N}(\sqrt{2mE})$$

where $A_{3N}(\sqrt{2mE})$ is the surface area of hypersphere of radius $r = \sqrt{2mE}$ and dimension $d = 3N$.

And so, we have the ideal gas partition function^a in canonical ensemble as

$$Z_{ideal}(N, V, T) = \frac{1}{N!} (V/\lambda^3)^N$$

Hard sphere exclusion Suppose each particle acts as a *hard sphere*, each occupying a volume $\zeta \ll V$, then

$$\int \prod_{i \leq N} d^3 \vec{q}_i \simeq V^N \left(1 - \frac{N}{2V} \zeta\right)^N$$

$$p(V - \frac{N\zeta}{2}) = Nk_B T$$

Ultra - relativistic Ideal Gas For this gas $pc \gg m_0 c^2$. So the total energy of the system is given by

$$E = \sqrt{p^2 c^2 + m_0^2 c^4} \gtrsim pc \implies 0 \leq \sum_i |\vec{p}_i| \lesssim E/c$$

We also assume a spherical symmetry in momentum phase space, so changing coordinates

$$d\Gamma_j^p \sim d^3 p_j \rightarrow 4\pi v_j^2 dv_j \quad (v_j = |\vec{p}_j|)$$

$$\therefore \int \prod_{i \leq N} d^3 \vec{p}_i = (4\pi)^N \int_{0 \leq \sum_i v_i \lesssim E/c} \prod_{i \leq N} v_i^2 dv_i = V_{3N}(E/c)$$

$V_{3N}(E/c) = \frac{(8\pi E^3/c^3)^N}{(3N)!}$ is the volume of hypersphere of radius E/c and $d = 3N$.

$\lambda = h/\sqrt{2\pi m k_B T}$ the average thermal wavelength

Canonical Ensemble

Construction This ensemble is defined to be a system (S) with no work done/particle exchange, and is characterised by fixed temperature T (in contact with reservoir R). A macrostate is specified by $\mathbf{M}(T, \vec{x})$. However, the entire $R \oplus S$ resembles a microcanonical ensemble with $E_{tot} = E_S + E_R$ constant.

Let $|n\rangle$ denote a (micro)state of S with energy E_n , then corresponding to each E_n , $\exists \Omega_S(E_n)$ number of microstates in S of equal energy (**degeneracy**). Hence, the total number of microstates of the combined ($\mu_S \otimes \mu_R$) is given by *sum over all states of S*-

$$\begin{aligned} \Omega_{R \oplus S}(E_{tot}) &= \sum_n \Omega_R(E_{tot} - E_n) \\ &\equiv \left(\sum_{E_n} \Omega_S(E_n) \Omega_R(E_{tot} - E_n) \right) \\ &\sim e^{S_R(E_{tot})/k_B} \sum_k e^{-\beta E_k} \end{aligned}$$

Thus the probability that system S sits in $|n\rangle$ state is given by (from *equal a-priori* assumption)

$$p(\mu_S = |n\rangle) \equiv p(n) = \frac{\Omega_R(E_{tot} - E_n)}{\Omega_{R \oplus S}(E_{tot})} = \frac{\Omega_S(E_n) \times e^{-\beta E_n}}{Z}$$

where $Z = \sum_k e^{-\beta E_k} = \sum_{E_k} \Omega_S(E_k) e^{-\beta E_k}$ is the *partition function* of system and $\beta = 1/k_B T$.

Useful properties about Z

1. $\langle E \rangle = -\frac{\partial \ln Z}{\partial \beta}$ (On a macroscopic scale: $\langle E \rangle \rightarrow E_{\max}$)
2. $C_V = \frac{1}{k_B T^2} \Delta E^2 = \frac{1}{k_B T^2} \frac{\partial^2 \ln Z}{\partial \beta^2}$
3. *Helmholtz Free energy*: $F(T, V, N) = -k_B T \ln Z \equiv E - TS$
4. Consider a system of N identical particles. Let Z_1 denote the partition function of a single particle. Then we can write the total function for the system as $Z_d = Z_1^N$ (if **distinguishable particles**) OR $Z_{nd} = Z_1^N / N!$ (if **indistinguishable particles**)
5. For a given hamiltonian $H(p, q)$ of a *classical* system of N identical and indistinguishable particles, and element $d\Gamma$ in the phase space corresponds to $d\Gamma / (N! h^{3N})$ distinct quantum states of the system. In that case,

$$Z(T, V, N) = \frac{1}{h^{3N} N!} \int e^{-\beta H(p, q)} \prod_{i \leq N} d^3 \vec{q}_i d^3 \vec{p}_i$$

Gibbs Entropy Here, we realise entropy to be a function of probability distribution!

$$\mathcal{S} = -k_B \sum_n p(n) \ln p(n)$$

1. *Microcanonical Ensemble*: For $p(n) = 1/\Omega(E, V, N)$, this reduces to famous **Boltzmann entropy** $\mathcal{S}(E) = k_B \ln \Omega(E)$
2. *Canonical Ensemble*: $p(n) = e^{-\beta E_n} / Z$, this reduces to $\mathcal{S}(T) = k_B \frac{\partial(T \ln Z(T))}{\partial T}$

In canonical ensemble, the expectation value of any quantity $f(p, q)$ is defined as

$$\langle f(p, q) \rangle = \frac{\int f(p, q) e^{-\beta H(p, q)} d\Gamma}{\int e^{-\beta H(p, q)} d\Gamma}$$

Virial Theorem

It states that

$$\left\langle X_i \frac{\partial H}{\partial X_j} \right\rangle = \delta_{ij} k_B T$$

where X_j 's are the generalised canonical coordinates of the system. Further it follows that if $\langle \sum_i p_i \dot{q}_i \rangle = -\langle \sum_i q_i \dot{p}_i \rangle = 3Nk_B T$. We assume when evaluated at endpoints $X_{j, \max}, X_{j, \min}$, then $H \rightarrow \infty \implies e^{-\beta H} \rightarrow 0$. This is because,

- if $X_j = q_j$, the potential energy blows up and so H , (for eg, at walls of the container).
- Similarly, if $X_j = p_j$, the kinetic energy would be infinite and so H .

Equipartition Theorem

In many physical situation, one can obtain the hamiltonian of a system, by a suitable canonical transformation as

$$\begin{aligned} \mathcal{H} &= \sum_j (A_j Q_j^2 + B_j P_j^2) \\ &= \sum_j \left(\frac{1}{2} \frac{\partial \mathcal{H}}{\partial Q_j} Q_j + \frac{1}{2} \frac{\partial \mathcal{H}}{\partial P_j} P_j \right) \\ \implies E \equiv \langle \mathcal{H} \rangle &= \sum_j \left(\frac{1}{2} \left\langle \frac{\partial \mathcal{H}}{\partial Q_j} Q_j \right\rangle + \frac{1}{2} \left\langle \frac{\partial \mathcal{H}}{\partial P_j} P_j \right\rangle \right) \\ &= \sum_j k_B T = f k_B T \end{aligned}$$

Where, f denotes the number of non-vanishing harmonic terms in $\mathcal{H}$. Hence, this theorem states that each term in $\mathcal{H}$ contributes to $1/2 \times k_B T$ to the total energy of the system. Similarly for a general Non-harmonic system $\mathcal{H}' = \sum_j (A_j Q_j^\delta + B_j P_j^\delta)$, where $\delta \in \mathbb{R}^+ - \{0\}$, per term contribution is $k_B T / \delta$.

Maxwell Boltzmann distribution

A single particle hamiltonian is $H(v, q) = mv^2/2$, where $\mathbf{p} = m\mathbf{v} \implies d^3p = m^3 d^3v$. Thus

$$Z_1 = \frac{m^3}{h^3} \int_{\text{box}} d^3q \int d^3v \exp\{-\beta mv^2/2\} \\ \sim \int_0^\infty dv v^2 \exp\{-\beta mv^2/2\}$$

Hence, the probability that the atom has speed between v and $v + dv$ is

$$f_{\text{Boltz}}(v)dv = \mathcal{N}v^2 \exp\{-\beta mv^2/2\}dv$$

where $\int_0^\infty dv f_{\text{Boltz}}(v) = 1$.

Some features of f_{Boltz} distribution are:

1. Most Probable speed

$$\left. \frac{df_{\text{Boltz}}}{dv} \right|_{v_{mp}} = 0 \implies v_{mp}^2 = \frac{2}{m\beta}$$

2. Average speed

$$\langle v \rangle = \int dv v f_{\text{Boltz}}(v) \implies \langle v \rangle^2 = \frac{8}{\pi m\beta}$$

3. RMS speed

$$\langle v^2 \rangle = \int dv v^2 f_{\text{Boltz}}(v) \implies \langle v^2 \rangle = \frac{3}{m\beta}$$

Diatomic Gas

Assumption: Rotation and vibration modes are independent of each other. The gas is non interacting

$$Z_1 = Z_{\text{trans}} Z_{\text{rot}} Z_{\text{vib}}$$

1. Translation

$$H = \frac{(p_x^2 + p_y^2 + p_z^2)}{2m} ; Z_{\text{trans}} = \frac{V}{\lambda^3}$$

2. Rotation

$$\mathcal{L} = \frac{I}{2}(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) \implies H = \dot{q}_i p_i - \mathcal{L} = \frac{p_\theta^2}{2I} + \frac{p_\phi^2}{2I \sin^2 \theta}$$

$$\therefore Z_{\text{rot}} = \frac{1}{h^2} \int d\theta d\phi dp_\theta dp_\phi e^{-\beta H_{\text{rot}}} = \frac{2I}{\beta \hbar^2}$$

3. Vibration

$$H = \frac{p_\zeta^2}{2m} + \frac{m\omega^2 \zeta^2}{2} ; Z_{\text{vib}} = \frac{k_B T}{\hbar \omega}$$

$$\text{For gas, } Z = \frac{Z_1^N}{N!} \implies E = (3 + 2 + 2) \times \frac{k_B T}{2}$$

Interacting Gas

As soon as we take into account of an interaction potential $U(r)$, the most general equation of state becomes

$$\frac{P}{k_B T} = \frac{N}{V} + B_2(T) \frac{N^2}{V^2} + B_3(T) \frac{N^3}{V^3} + \dots$$

Our goal is to compute $B_j(T)$ s(also called *virial coefficients*). We have

$$H = \sum_{i=1}^N \frac{p_i^2}{2m} + \sum_{i>j} U(r_{ij})$$

where $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ is the separation between particles. The restriction $i > j$ on the final sum ensures that we sum over each pair of particles exactly once. The partition function is then

$$Z(N, V, T) = \frac{1}{N! h^{3N}} \int d\Gamma e^{-\beta H} \\ = \frac{1}{N! \lambda^{3N}} \int \prod_i d^3 r_i e^{-\beta \sum_{j<k} U(r_{jk})} \\ = \frac{1}{N! \lambda^{3N}} \int \prod_i d^3 r_i \prod_{j>k} (1 + f_{jk})$$

where we define *Mayer f function*

$$f(r) = e^{-\beta U(r)} - 1$$

which behaves *nicely* at the extreme points!. Now changing variables to center of mass coordinates and separation, we can evaluate each pair of terms separately. Massaging enough, gives the final expression

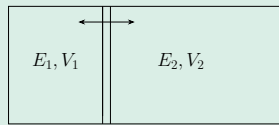
$$Z(N, V, T) = Z_{\text{ideal}} \left(1 + \frac{N}{2V} \int d^3 r f(r) + \dots \right)^N$$

Recalling that, $F = -k_B T \ln Z$, we calculate $P = -\left(\frac{\partial F}{\partial V}\right)$ to obtain the **equation of state**

$$\frac{PV}{Nk_B T} = \left(1 - \frac{N}{2V} \int d^3 r f(r) + \dots \right)$$

Gibbs Canonical Ensemble

Motivation Unlike, microcanonical and canonical ensemble where we held the V, N fixed. Here we vary V . In the Gibbs ensemble we have two systems in equilibrium that can exchange *energy* and *volume*. Exchanging volume just means we have a moveable partition in between them. So the total volume ($V_{tot} = V_1 + V_2$) is conserved. Let the total energy be $E_{tot} = E_1 + E_2$.



The entropy of the system-

$$S_{12}(E_{tot}, V_{tot}, E, V) = S_1(E, V) + S_2(E_{tot} - E, V_{tot} - V)$$

Then,

Equilibrium $\implies$ Maximising Entropy

- Thermal Equilibrium: $\left(\frac{\partial S_{12}}{\partial E}\right)_{V, N} \Big|_{E=E_1} = 0 \implies T_1 = T_2$
- Mechanical Equilibrium: $\left(\frac{\partial S_{12}}{\partial V}\right)_{E, N} \Big|_{V=V_1} = 0 \implies p_1/T_1 = p_2/T_2 \implies p_1 = p_2$.

Construction We are thus led to define a generalized canonical ensemble in which the internal energy changes by the addition of both heat and work. The macrostates $\mathbf{M}(T, \mathbf{J})$, are specified in terms of the external temperature and forces acting on the system. Including the work done against the forces, the energy of the combined system that includes these elements is $H - \mathbf{J} \cdot \mathbf{x}$. Thus the microstates μ_S of the combined system occur with the probability

$$p(\mu_S, \mathbf{x}) = \frac{e^{-\beta(H(\mu_S) - \mathbf{J} \cdot \mathbf{x})}}{\mathcal{Z}(T, N, \mathbf{J})}$$

where the *Gibbs Partition Function* is

$$\mathcal{Z}(T, N, \mathbf{J}) = \sum_{\mu_S, \mathbf{x}} e^{-\beta(H(\mu_S) - \mathbf{J} \cdot \mathbf{x})}$$

- the expectation value of the coordinates is, $\langle \mathbf{x} \rangle = k_B T \frac{\partial \ln \mathcal{Z}}{\partial \mathbf{J}}$, where $\mathbf{x} = -\frac{\partial G}{\partial \mathbf{J}}$.
- *Gibbs free energy* defined as $G \equiv E - TS - \mathbf{x} \cdot \mathbf{J}$, is obtained from $G(N, T, \mathbf{J}) = -k_B T \ln \mathcal{Z}$.
- *Enthalpy* $\mathcal{H} \equiv E - \mathbf{x} \cdot \mathbf{J} = -\frac{\partial}{\partial \beta} \ln \mathcal{Z}$
- heat capacities at constant force $C_{\mathbf{J}} = \frac{\partial \mathcal{H}}{\partial T}$

Ideal gas in isobaric ensemble It is described by the microstate $M(N, T, P)$. The probability of a particular microstate $\mu \equiv \{\vec{p}_i, \vec{q}_i\}$, in volume V is given by the probability ($\vec{q}_i \in V$),

$$p(\mu, V) = \frac{1}{\mathcal{Z}} \exp \left[-\beta \sum_{i=1}^N \frac{p_i^2}{2m} - \beta PV \right]$$

Thus,

$$\begin{aligned} \mathcal{Z} &\sim \int_0^\infty dV e^{-\beta PV} \int d\Gamma_{\text{id}} e^{-\beta \sum_i p_i^2 / 2m} \\ &\sim \frac{1}{(\beta P \lambda^3)^N} \end{aligned}$$

Spins in external B-field

$$\mathcal{Z}(N, T, B) = \text{Tr} [\exp (-\beta(H - \mathbf{B} \cdot \mathcal{M}))]$$

energy E_n becomes,

$$p(\mu_S = |n\rangle) = \frac{e^{-\beta(E_n - \mu N_n)}}{Z_G}$$

where Z_G is the Grand Canonical Partition Function

$$Z_G(T, \mu, V) = \sum_n e^{-\beta(E_n - \mu N_n)}$$

- $\mu \equiv \left(\frac{\partial E}{\partial N}\right)_{S, V} = -T \left(\frac{\partial S}{\partial N}\right)_{E, V}$
- $\langle E \rangle - \mu \langle N \rangle = -\frac{\partial}{\partial \beta} \ln Z_G$
- $\Delta N^2 = \frac{1}{\beta^2} \frac{\partial^2}{\partial \mu^2} \ln Z_G$
- **Fugacity** is denoted by $z = e^{\beta \mu}$

Grand Canonical Potential

$$\Phi \equiv F - \mu N$$

Φ is a legendre transform of F from variable μ to N .

$$d\Phi = -SdT - PdV - Nd\mu \implies \Phi \sim \Phi(T, V, \mu)$$

$$\therefore \Phi = -k_B T \ln Z_G = -PV$$

The ideal gas in GCE Recall,

$$Z_{\text{ideal}}(N, V, T) = \frac{1}{N!} \left(\frac{V}{\lambda^3} \right)^N$$

Then,

$$Z_G(\mu, V, T) = \sum_{N=0}^{\infty} e^{\beta \mu N} Z_{\text{ideal}}(N, V, T) = \exp \left[\frac{e^{\beta \mu} V}{\lambda^3} \right]$$

In the thermodynamic limit ($N \sim 10^{23}$)

$$N \simeq \langle N \rangle = \frac{1}{\beta} \frac{\partial Z_G}{\partial \mu} \implies \mu = \checkmark$$

For consistency of classical calculations, we must have $\lambda^3 \ll \sigma^a$

^a $\sigma = V/N$ is the average volume occupied by all the particles

Grand Canonical Ensemble

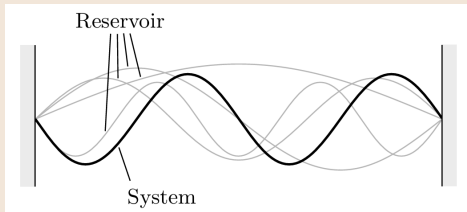
Construction In many real life systems, N is not fixed. So its convenient to associate a chemical potential μ . An ensemble where N changes but the total number of particles is conserved among system is called the grand canonical ensemble (gce). In such scenario, we require the reservoirs to sit at fixed chemical potential and temperature. The resulting macrostate is defined by $\mathbf{M}(T, \mu, \mathbf{x})$.

Thus, the probability of finding a state $|n\rangle$ with N_n particles and

Quantum Stat Mech

In quantum stat mech, we work in the *grand canonical ensemble*. We consider a system of particles to be a *quantum gas* if $V/N \approx v_Q = \lambda^3$, (where $\lambda = h/\sqrt{2\pi m k_B T}$ the thermal wavelength) i.e the wavefunctions begin to touch and overlap.

Construction Our "system" will consist of a single-particle state or a *spatial wavefunction*. The "reservoir" consists of all the other possible single-particle states.



$\therefore$ The probability of the state (of energy ϵ) being occupied by n particles is:

$$P(n) = \frac{e^{-\beta(n\epsilon - n\mu)}}{Z_G}$$

where the grand partition function $Z_G = \sum_n e^{-n\beta(\epsilon - \mu)}$.

Note that, if the grand partition function for a single-particle state $|r\rangle$ is $Z_{G,r}$ then the full grand partition function for the *system* is $Z_G = \prod_r Z_{G,r}$.

Occupancy The average number of particles in a state is defined as $\bar{n} = \sum_n nP(n)$

Planck distribution

Density of states (Method-1) Consider a gas of photons in a box of volume V . Let the degeneracy factor be g_p (for photons $g_p = 2$

since it has two possible polarizations). The total number of microstates of this single-particle state will be

$$g_p \times \int \frac{d^3\mathbf{q}d^3\mathbf{p}}{h^3} = \dots = g_p \times \frac{4\pi \cdot V}{h^3} \int dp p^2$$

Now, for relativistic particles moving in $d = 3 + 1$ spacetime dimensions have kinetic energy

$$\epsilon^2 = p^2 c^2 + m^2 c^4 \implies p^2 = \frac{1}{c^2}(\epsilon^2 - m^2 c^4)$$

Substituting and changing the variable, we obtain

$$\dots = \int g(\epsilon) d\epsilon$$

we get the density of states as,

$$g(\epsilon) = g_p \times \frac{4\pi V}{(hc)^3} \epsilon \sqrt{\epsilon^2 - m^2 c^4}$$

For massless particles like phonons ($g_p = 3$) and photons ($g_p = 2$)

$$g(\epsilon) d\epsilon = \frac{8\pi V}{(hc)^3} \epsilon^2 d\epsilon \iff g(\omega) d\omega = \frac{V \omega^2}{c^3 \pi^2} d\omega$$

where we have used the dispersion relation $\epsilon = \hbar\omega$ for photons. The quantity $g(\epsilon) d\epsilon$ should be viewed as a measure which we can integrate over any function $f(E)$ of our choosing.

NOTE that, because photon number is not conserved, we are unable to define a chemical potential for photons.

Partition Function We'll start by looking at photons with a definite frequency ω . A state with n such photons has energy $\epsilon_n = n\hbar\omega$. So, the partition function for photons at fixed frequency is

$$Z_\omega = \sum_{n=0}^{\infty} e^{-\beta\epsilon_n} = \sum_{n=0}^{\infty} e^{-\beta n\hbar\omega} = \frac{1}{1 - e^{-\beta\hbar\omega}}$$

We now need to sum over all possible frequencies.

$$Z = \prod_{\omega} Z_{\omega}^{(g(\omega))}$$

$$\log Z = \sum_{\omega} g(\omega) \log(Z_\omega) \simeq \int d\omega g(\omega) \log(Z_\omega) \checkmark$$

This leads us to the average energy of the system as

$$U = -\frac{\partial}{\partial\beta} \log Z \equiv \int_0^\infty d\omega (\hbar\omega) \bar{n}_{Pl}(\omega) g(\omega) = \int_0^\infty d\omega U(\omega)$$

Comparing we get the particle number distribution

$$\bar{n}_{Pl}(\omega, T) = \frac{1}{1 - e^{-\beta\hbar\omega}}$$

and the **planck distribution** as

$$U(\omega) d\omega = \frac{V\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\beta\hbar\omega} - 1} d\omega$$

Wien's Displacement Law The wavelength of peak energy is $\propto 1/T$

$$\frac{dU(\omega)}{d\omega} = 0 \implies \omega_{\max} = 2.822 \frac{k_B T}{\hbar} \implies \boxed{\lambda_m T(K) = b \simeq 0.0028}$$

Integrating, we can find the energy density $\mathcal{E} = U/V \propto T^4$.

Stefan Boltzmann's Law Now, the energy flux (f_ϵ) (by considering the photon to be emitting in only one direction i.e normally away from the object) is then

$$L = \mathcal{E} \times \frac{1}{4\pi} \int_0^{2\pi} d\phi \int_0^{\pi/2} d\theta \sin\theta (c \cos\theta) = \mathcal{E} \frac{c}{4}$$

$$\implies f_\epsilon = \sigma T^4$$

where, $\sigma = \frac{\pi^2 k_B^4}{60 h^3 c^2} \approx 5.67 \times 10^{-8} \text{ Js}^{-1} \text{ m}^{-2} \text{ K}^{-4}$.

Rayleigh Jeans Law In the regime $\hbar\omega \ll k_B T$ (radio wavelengths), we have

$$U(\omega) \simeq \frac{V}{\pi^2 c^3} \omega^2 k_B T$$

Radiation Pressure Recall, $\frac{PV}{k_B T} = \ln Z \implies P = \frac{E}{3V}$.

Phonon Theory

EM Waves $\xrightarrow{\text{Q Mech}}$ Photons
 Sound Waves (in solid) $\xrightarrow{\text{Debye Model}}$ Phonons

In contrast to photons, a phonon has $g_p = 3$ polarization states (2T + 1L). The density of states is

$$g(\omega) = \frac{3V}{2\pi^2 c_s^3} \omega^2$$

where the dispersion relation follows as $E = \hbar\omega = \hbar kc_s$.

Debye Frequency Unlike photons, there exists a minimum wavelength $\lambda_{\min} \sim (V/N)^{1/3}$ with which phonons can vibrate corresponding to the finite non-zero distance between atoms. This maximum allowed frequency is called debye frequency (ω_D). Similarly, we define *debye temperature* $T_D = \hbar\omega_D/k_B$ to characterize solids.

Debye Theory According to this theory, the total number of one-phonon states is equal to the total number of degrees of freedom: it is the number of things you can excite.

$$\int_0^{\omega_D} g(\omega) d\omega = 3N \implies \omega_D = \left(\frac{6\pi^2 N}{V} \right)^{1/3} c_s$$

Heat Capacity of Solids

Like photons, the number of phonons is not conserved. So, we can write the total partition as

$$\log Z_{\text{phonon}} = \int_0^{\omega_D} d\omega g(\omega) \log Z_\omega$$

The total energy in sound waves is therefore,

$$E = -\frac{\partial}{\partial \beta} \log Z_{\text{phonon}} = \frac{3V}{2\pi^2 (\hbar c_s)^3} (k_B T)^4 \int_0^{T_D/T} dx \frac{x^3}{e^x - 1}$$

where $x = \beta \hbar \omega$.

CASE-1 ($T \ll T_D$) We can safely replace the upper limit as ∞ . In which case, the integral = $\pi^4/15$. Hence, in *low* temperature regime

$$C_V = N k_B \frac{12\pi^4}{5} \left(\frac{T}{T_D} \right)^3 \propto T^3$$

CASE-2 ($T \gg T_D$) We integrate only over small values of x . Hence, we neglect the higher orders of the integrand.

$$\int_0^{T_D/T} dx \frac{x^3}{e^x - 1} = \int_0^{T_D/T} dx (x^2 + \dots) = \frac{1}{3} \left(\frac{T_D}{T} \right)^3 + \dots$$

This ensures that the energy grows linearly with T . Thus, the heat capacity is constant!

$$C_V = \frac{V k_B^4 T_D^3}{2\pi^2 \hbar^3 c_s^3} = 3N k_B$$

This is called the famous **Dulong-Petit law**.

Drawbacks of the model Problem lies in the assumption of *dispersion relation* as $E(k) = \hbar\omega_k = \hbar c_s k$. In reality, phonons should be worked in a periodic lattice (sep by a). If we approximate the bonds joining the atoms as harmonic oscillators we get,

$$\omega_k = 2\sqrt{\frac{\alpha}{m}} \left| \sin \left(\frac{ka}{2} \right) \right|$$

where, α is a parameter governing the strength of the bonds between atoms, $c_s = a\sqrt{\alpha/m}$.

Bose Systems

Bosons are

- Integer spin particles
- Symmetric: $\psi(\vec{r}_1, \vec{r}_2) = \psi(\vec{r}_2, \vec{r}_1)$

- photon, He^4 nuclei, gravitons, higgs boson
- Unlike fermions, a *single-particle state* can be occupied by any number of particles.

Grand Partition Function For a single-particle state

$$Z_G = \sum_{n=0}^{\infty} e^{-n\beta(\epsilon-\mu)} = \frac{1}{1 - ze^{-\beta\epsilon}}$$

However, it is to be noted for convergence we require $\epsilon > \mu$.

$$\mathcal{Q} = \prod_{\epsilon} Z_G(\epsilon)$$

Bose- Einstein Distribution The occupancy is calculated as

$$\begin{aligned} \bar{n} &= 0 \cdot P(0) + 1 \cdot P(1) + \dots \\ &= \sum_{n=0}^{\infty} n \frac{e^{-n\beta(\epsilon-\mu)}}{Z_G} \\ &= -\frac{\partial}{\partial x} \log Z_G \end{aligned}$$

where, $x = \beta(\epsilon - \mu)$. Hence we get,

$$\implies \bar{n}_{\text{BE}}(\epsilon, T) = \frac{1}{e^{\beta(\epsilon-\mu)} - 1} = \frac{1}{z^{-1}e^{\beta\epsilon} - 1}$$

Ideal Bose Gas

As calculated in the section of [Fermi Systems](#), the density states is

$$g(\epsilon) = \frac{2V\beta^{3/2}}{\lambda^3 \sqrt{\pi}} \sqrt{\epsilon}$$

This makes it easy for us to calculate the total number of particles,

$$N = \int_0^{\infty} d\epsilon \bar{n}_{\text{BE}}(\epsilon) g(\epsilon)$$

However, this suggests a dependence of N on μ, T , but this is not so! as it is observed to be *constant* experimentally. Thus, our goal is invert this expression to obtain $\mu(N, T)$. Then, put this in $U = U(\mu, T)$ to get the *equation of state*! Recall,

$$U = \int_0^{\infty} d\epsilon \epsilon \bar{n}_{\text{BE}}(\epsilon) g(\epsilon)$$

$$\frac{PV}{k_B T} = \ln \mathcal{Q} = - \int_0^{\infty} d\epsilon g(\epsilon) \ln(1 - ze^{-\beta\epsilon})$$

Integrating by parts, we get

$$PV = 2U/3$$

CASE-1: $z = e^{\beta\mu} \ll 1$ (**High Temperature**) The equation of state is

$$PV = N k_B T \left(1 - \frac{\lambda^3 N}{4\sqrt{2}V} + \dots \right)$$

Hence, the first order quantum correction to a bose gas increases the pressure at high temperature.

CASE-2: $z \rightarrow 1$ (**Low Temperature**) Since, for convergence of $\mathcal{Q}$ we require $\mu < 0$ ^a. So $0 < z < 1$. For High temp $z \rightarrow 0$, thus we expect $z \rightarrow 1$ in low temp regime.

As we lower the temp, the ground state gets occupied by macroscopic number of particles. i.e

$$\langle n_0 \rangle = \frac{1}{z^{-1} - 1} \sim N$$

Also note, as $g(\epsilon) \propto \sqrt{\epsilon}$, clearly the integral $\int_0^{\infty} d\epsilon \bar{n}_{\text{BE}}(\epsilon) g(\epsilon)$ vanishes at low temp regime. (indicating an unrealistic vanishing of N despite the fact that it is constant). So it is wise to add this term to correct for the total number of particles.

$$N = \frac{V}{\lambda^3} g_{3/2}(z) + \frac{1}{z^{-1} - 1}$$

This means that as $T \rightarrow 0$, z hovers around $1 - \frac{1}{N}$.

Critical Temperature (T_c) Rewriting the above equation,

$$\begin{aligned}\lambda^3 \frac{\langle n_0 \rangle}{V} &= \lambda^3 \frac{N}{V} - g_{3/2}(z) \geq 0 \\ \implies \lambda^3 \frac{N}{V} &\geq g_{3/2}(z) \quad \forall z \\ \implies \lambda^3 \frac{N}{V} &\geq g_{3/2}(1) = \zeta(3/2)\end{aligned}$$

We define $T = T_c$ to be the temperature at which

$$\lambda^3(T_c) \frac{N}{V} = \zeta(3/2)$$

This defines a transition between a gas phase and a *condensed* phase. Thus

$$T_c = \left(\frac{2\pi\hbar^2}{mk_B} \right) \left(\frac{1}{\zeta(3/2)} \frac{N}{V} \right)^{2/3} \propto N^{2/3}$$

At temperatures $T < T_c$ ^b, a macroscopic number of atoms discard their individual identities and merge into a communist collective, a single quantum state so large that it can be seen with the naked eye. This is known as the **Bose-Einstein condensate**.

Equation of State

$$\begin{aligned}\ln \mathcal{Q} &= - \sum_r \ln(1 - ze^{-\beta\epsilon}) \\ &= - \ln(1 - z) - \sum_{r \neq 0} \ln(1 - ze^{-\beta\epsilon}) \\ &\simeq - \ln(1 - z) - \int d\epsilon g(\epsilon) \ln(1 - ze^{-\beta\epsilon}) \\ &= - \ln(1 - z) + \frac{V}{\lambda^3} g_{5/2}(z)\end{aligned}$$

$$\therefore \frac{P}{k_B T} = \frac{1}{\lambda^3} g_{5/2}(z) - \frac{\ln(1 - z)}{V}$$

Heat Capacity Recall, $\frac{U}{V} = \frac{3}{2} \frac{k_B T}{\lambda^3} g_{5/2}(z)$. So,

$$\frac{C_V}{V} = \frac{1}{V} \frac{dU}{dT} = \frac{15k_B}{4\lambda^3} g_{5/2}(z) + \frac{3}{2} \frac{k_B T}{\lambda^3} \frac{dg_{5/2}}{dz} \frac{dz}{dT}$$

Analysing each cases, leads us to conclude

$$C_V = \begin{cases} \frac{15}{4} \frac{V k_B}{\lambda^3} \zeta(5/2), & T < T_c \\ \frac{15}{4} \frac{V k_B}{\lambda^3} g_{5/2}(z) - b \left(\frac{T - T_c}{T_c} \right), & T \geq T_c \end{cases}$$

It is to be noted that, Phase transitions with their associated discontinuities can only arise in strictly infinite systems. There are no phase transitions in finite systems

^asince $\mu < \epsilon_s$ for all states and for ground state $\epsilon_s = \mu$

^b $z = 1$ (and remains) in the thermodynamic limit

Fermi Systems

Fermions are

- 1/2 integer spin particles
- Antisymmetric: $\psi(\vec{r}_1, \vec{r}_2) = -\psi(\vec{r}_2, \vec{r}_1)$
- electrons, protons, neutrons, He³ nuclei.
- Introvert: Two identical fermions can **never** occupy the same state (so occupation number n_r can be 0 or 1). This is called **Pauli Exclusion Principle**.

Fermi-Dirac distribution The occupancy of a system is calculated as

$$\bar{n} = 0 \cdot P(0) + 1 \cdot P(1)$$

$$\implies \bar{n}_{\text{FD}}(\epsilon, T) = \frac{1}{1 + e^{\beta(\epsilon - \mu)}} = \frac{1}{1 + z^{-1} e^{\beta\epsilon}}$$

Grand Partition Function For a single-particle state

$$Z_G = 1 + e^{-\beta(\epsilon - \mu)} = 1 + ze^{-\beta\epsilon}$$

Density of states (Method-2) The following derivation can be followed for any type of particle distribution. Consider a ideal gas of *non-interacting non-relativistic* fermions (at $T = 0$!), enclosed in a volume $V = L^3$ (if say, a cube). As described earlier, the energy states are actually localised in the momentum space. Since, there are no interactions, each energy eigenstate of a particle is a plane wave ($p_i = \hbar k_i$). So, the allowed energy for a particle can take the form

$$\epsilon = \frac{\hbar^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2) \propto |\vec{n}|^2 \quad ; n_i \in \mathbb{Z}^+ \cup \{0\}$$

Let $g_s = 2s + 1$ denote the **degeneracy** of the fermions. The total energy of all the fermions is

$$\begin{aligned}U &= \sum_{n_x} \sum_{n_y} \sum_{n_z} \epsilon(\vec{n}) \approx g_s \int \int \int d^3 n_i \epsilon(n_1, n_2, n_3) \\ &= g_s \int_0^{n_{\max}} dn n^2 \epsilon(n) \times \int_0^{\pi/2} d\phi \times \int_0^{\pi/2} \sin \theta d\theta\end{aligned}$$

Changing variables : $\epsilon = \frac{\hbar^2}{8mL^2} n^2 \rightarrow dn = (\dots) d\epsilon$

$$U = \int_0^{\epsilon_F} g(\epsilon) d\epsilon$$

Thus, we define $g(\epsilon)$ as the number of single particle states per unit energy or *density of states*!

$$g(\epsilon) = g_s \frac{\pi(8m)^{3/2}}{4\hbar^3} V \sqrt{\epsilon} = g_s \frac{3N}{4\epsilon_F^{3/2}} \sqrt{\epsilon} = g_s \frac{2V\beta^{3/2}}{\lambda^3 \sqrt{\pi}} \sqrt{\epsilon}$$

For Relativistic system $\epsilon^2 = \hbar^2 k^2 c^2 + m^2 c^4$. Repeating the above steps, one can obtain

$$g(\epsilon) = g_s \frac{V\epsilon}{4\pi^2 \hbar^3 c^3} V \sqrt{\epsilon^2 - m^2 c^4}$$

Fermi Quantities Now, After filling all the particles, all the lattice points will occupy an octant of a sphere in the **n** space. (Think!) Denote it's radius by $n_{\max}$.

- **Fermi Energy** is the highest energy of an occupied fermion, $\epsilon_F = \frac{\hbar^2}{8mL^2} n_{\max}^2$
- **Fermi momentum** is defined as $\hbar k_F = \sqrt{2m\epsilon_F}$. All the states with $|\vec{k}| = k_F$ form the **Fermi Surface**, while those with $|\vec{k}| \leq k_F$ are called *Fermi Sea / Sphere*.
- **Fermi Temperature** $T_F = \epsilon_F/k_B$. Temperatures $T \ll T_F$ describe Low or small non-zero Temperature, while $T > T_F$ are called high temperature

Now, the total number of occupied states is also given by

$$N = g_s \times \frac{1}{8} \times \frac{4}{3} \pi n_{\max}^3$$

Inverting this expression for $n_{\max}$, we get

$$\epsilon_F = \frac{\hbar^2}{8m} \left(\frac{6N}{\pi g_s V} \right)^{2/3}$$

CASE-1 : $T = 0$ [Degenerate]

$$\bar{n}_{\text{FD}} = \begin{cases} 1, & \text{if } \epsilon \leq \epsilon_F \equiv \mu \\ 0, & \text{if } \epsilon > \mu \end{cases}$$

As the number of particles N (or occupied states) is fixed, at $T=0$ the chemical potential $\mu \equiv \epsilon_F$, the energy required to add one more fermion to the system.

- $N = \int_0^{\epsilon_F} d\epsilon g(\epsilon)$
- Average energy of the system, $U = \int_0^{\epsilon_F} d\epsilon \epsilon g(\epsilon) = 3N\epsilon_F/5$
- Equation of state

$$PV = \frac{1}{\beta} \ln Z_G = \frac{1}{\beta} \int_0^{\epsilon_F} d\epsilon g(\epsilon) \ln(1 + ze^{-\beta\epsilon}) = 2U/3$$

$$\implies P = \frac{2}{5} N \epsilon_F / V \text{ Degeneracy Pressure !!}$$

CASE-2 : $T > T_F$ [High Temperature]

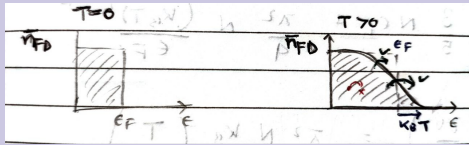
The equation of state is

$$PV = Nk_B T \left(1 + \frac{\lambda^3 N}{4\sqrt{2}g_s V} + \dots \right)$$

Hence, the first order quantum correction to a fermi gas increases the pressure. For $z \ll 1$, the equation of state reduces to that of a classical gas.

CASE-3 : $T \ll T_F$ [Low Temperature]

The transition happens as



In this case the chemical potential, μ , is the point where the probability of a state being occupied is exactly 1/2. As T increases from 0, overall all particles acquire thermal energy $\sim k_B T$. Thus only the particles, at the fermi surface has place to move and occupy higher states (above ϵ_F). This leads to the decrease in μ (as N is to be kept fixed).

Total number of particles

$$\begin{aligned} N &= \int_0^\infty d\epsilon \bar{n}_{FD}(\epsilon) g(\epsilon) \\ &= \frac{g_s V}{\lambda^3} f_{3/2}(z) \\ &= \frac{2}{3} g_0 \mu^{3/2} + \frac{1}{4} g_0 (k_B T)^2 \mu^{-1/2} \cdot \frac{\pi^2}{3} + \dots \\ \Rightarrow \frac{N}{V} &= \frac{g_s}{6\pi^2 \hbar^3} (2m\mu)^{3/2} \left(1 + \frac{\pi^2}{8} \left(\frac{k_B T}{\mu} \right)^2 + \dots \right) \end{aligned}$$

where $g_0 = 3N g_s / (4\epsilon_F^{3/2})$.

Chemical Potential

$$\mu = \epsilon_F \left(1 - \frac{\pi^2}{12} \left(\frac{k_B T}{\epsilon_F} \right)^2 + \dots \right)$$

Total energy

$$\begin{aligned} U &= \int_0^\infty d\epsilon \epsilon \bar{n}_{FD}(\epsilon) g(\epsilon) \\ &= \frac{3V g_s}{2\beta \lambda^3} f_{5/2}(z) \\ \Rightarrow \frac{U}{V} &= \frac{g_s}{10\pi^2 \hbar^3} (2m)^{3/2} \mu^{5/2} \left(1 + \frac{5\pi^2}{8} \left(\frac{k_B T}{\mu} \right)^2 + \dots \right) \\ \Rightarrow U &= \frac{3}{5} N \epsilon_F + \frac{\pi^2}{4} N \frac{(k_B T)^2}{\epsilon_F} + \dots \end{aligned}$$

Equation of State From grand canonical ensemble,

$$\begin{aligned} \frac{PV}{k_B T} &= \ln Z_G \equiv \int_0^\infty d\epsilon g(\epsilon) \bar{n}_{FD}(\epsilon) \ln(1 + z e^{-\beta \epsilon}) \\ &= g_s \times \frac{V}{h^3} \int_0^\infty dp 4\pi p^2 \ln(1 + z e^{-\beta p^2/2m}) \\ &= \dots \text{Do some massaging} \\ \Rightarrow \frac{P}{k_B T} &= \frac{g_s}{\lambda^3} f_{5/2}(z) \Rightarrow PV = \frac{2}{3} U ! \\ &= \frac{g_s}{\lambda^3} \sum_{u=1}^\infty \frac{(-1)^{u+1}}{u^{5/2}} z^u \end{aligned}$$

Heat Capacity

$$C_V = \left. \frac{\partial E}{\partial T} \right|_{N,V} = N k_B \frac{\pi^2}{2} \frac{T}{T_F} \propto T$$

Some Applications

1. **Heat Capacity of Metals** The vibrations of the lattice gives a phonon contribution which goes as T^3 . And if the conduction electrons act as an ideal gas, they give a linear term (due to quantum effects)

$$C_V = \gamma T + \alpha T^3$$

2. **White Dwarfs**

3. **Neutron Stars**

4. **Pauli Paramagnetism** In the presence of magnetic field, the new energy of each state becomes $\epsilon + \mu_B B s$, where $s = 1$ or -1 depending upon $|\uparrow\rangle$ or $|\downarrow\rangle$.

- $N = N_1 + N_2$
- $\frac{N_{\uparrow,\downarrow}}{V} = 1/\lambda^3 \cdot f_{3/2}(ze^{\mp \mu_B B})$
- At high temp, and suitably low magnetic fields $f_{3/2}(z) \approx z$.
- Total magnetisation, $\mathcal{M} = -\frac{\partial E}{\partial B} = -\mu_B (N_\uparrow - N_\downarrow) \simeq \mu_B N \tanh(\beta \mu_B B)$
- Magnetic Susceptibility, $\chi = \frac{\partial \mathcal{M}}{\partial B} \xrightarrow{B \rightarrow 0} \frac{N \mu_B^2}{k_B T} \propto 1/T$ [Curies Law!]
- However at low temperature $\mathcal{M} \approx \mu_B^2 B g(\epsilon_F) \Rightarrow \chi = \mu_B^2 g(\epsilon_F) > 0$. [Thus Curies Law is no longer obeyed, and saturates to constant. This is the rise of paramagnetism.]

5. **Landau Diamagnetism** Including the factor of 2 from electron spin, we can compute the grand partition function for non-interacting electrons in a magnetic field,

$$\log \mathcal{Z} = \frac{L}{2\pi} \int dk_z \sum_{n=0}^\infty \frac{2L^2 B}{\Phi_0} \log \left[1 + z \exp \left(-\frac{\beta \hbar^2 k_z^2}{2m} - \beta \hbar \omega_c \left(n + \frac{1}{2} \right) \right) \right]$$

So,

$$\mathcal{M} = \frac{1}{\beta} \frac{\partial}{\partial B} \log \mathcal{Z} = -\frac{\mu_B^2}{3} g(\epsilon_F) B$$

Notice, as compared to the Pauli contribution, it has an opposite sign. Substances whose magnetization is opposite to the magnetic field are called *diamagnetic*: they are repelled by applied magnetic fields.

Miscellaneous

Hamiltons Equations of Motion

∂H/∂p_i = q̇_i ; ∂H/∂q_i = -ṗ_i

Poisson Brackets

{A, B} = ∑_i ∂A/∂q_i ∂B/∂p_i - ∂A/∂p_i ∂B/∂q_i

where A(p, q, t) and B(p, q, t). Integrals/Constants of motion implies {A, H} = 0.

HUP

ΔxΔp ≡ ΔEΔτ ≥ ħ/2

Gamma Function For n ∈ ℤ+

Γ(n) = (n - 1)!

For z ∈ ℂ with Re(z) > 0

Γ(z) = ∫_0^∞ e^{-t} t^{z-1} dt

Useful: Γ(1/2) = √π, Γ(3/2) = √π/2

Stirling’s Formula For N >> 1 in the macroscopic limit

ln N! ≃ N ln N - N + ⋯ + O(1/N)

Hypersphere Consider an hyphersphere of radius r and dim d,

- 1. Surface Area

A_d(r) = (2π^{d/2} / (d/2 - 1)!) r^{d-1}

- 2. Volume

V_d(r) = (π^{d/2} / (d/2)!) r^d

Bose-Einstein Functions or Polylogarithms

Li_ν(z) or g_ν(z) ≡ 1/Γ(ν) ∫_0^∞ dx x^{ν-1} / (z^{-1}e^x - 1) = z + z^2/2^ν + z^3/3^ν + ⋯

Note that, g_n(1) = ζ(n), where, ζ is called the Reimann zeta function. [ζ(3/2) ≃ 2.612]

Fermi Dirac Functions

-Li_ν(-z) or f_ν(z) ≡ 1/Γ(ν) ∫_0^∞ dx x^{ν-1} / (z^{-1}e^x + 1)

At low temperature or large z, asymptotic expression is

f_ν(z) = (log z)^ν / Γ(ν + 1) (1 + π^2 ν(ν - 1) / (6(log z)^2) + ⋯)

Funny Integrals

- 1.

∫_0^∞ dx x^2 e^{-kx^2} = -∂/∂k (∫_0^∞ dx e^{-kx^2}) = -∂/∂k (1/2 √(π/k))

Useful Expansions

- ln(1 + x) = ∑_{n=1}^∞ (-1)^{n+1} x^n / n = x - x^2/2 + x^3/3 - ⋯